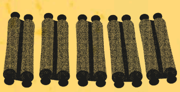
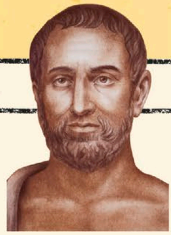
BELIEVED THERE WERE **5 ELEMENTS, INCLUDING 1** “HEAVENLY ELEMENT,” **QUINTESSENCE**

IDENTIFIED **500** LIVING SPECIES AND **11 GENERA** OF ANIMALS



31 OF ARISTOTLE’S **200 WORKS** SURVIVE

THEOPHRASTUS



A student and close colleague of Aristotle’s, Theophrastus was his successor at the Lyceum.

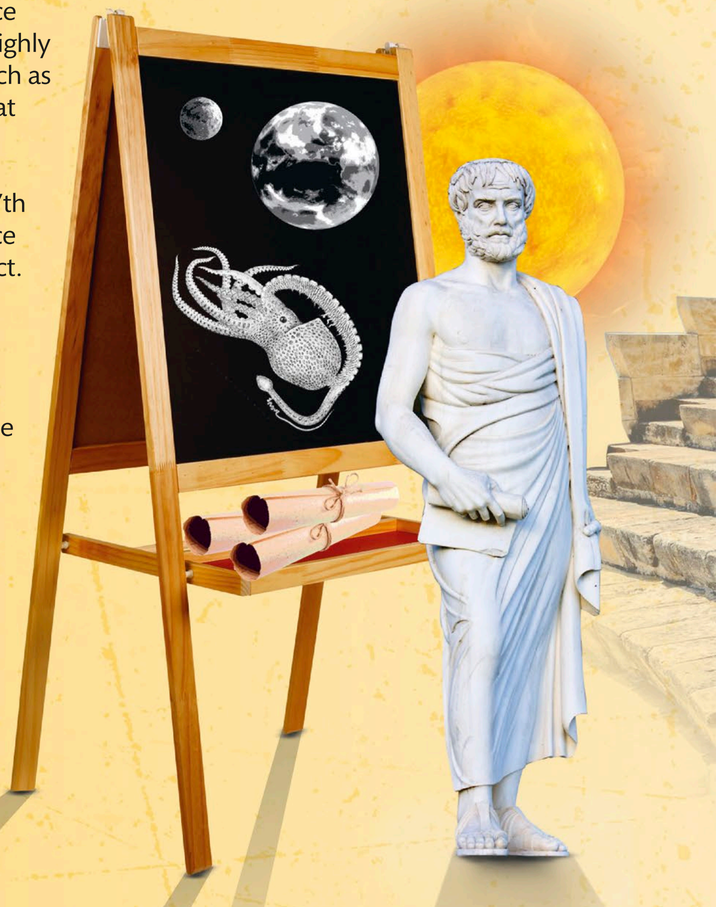
Like Aristotle, Theophrastus (c.371–287 BCE) had wide-ranging knowledge, but his special interest was botany. He described more than 500 species of plant, identified the role of leaves in plant nutrition, and was the first to use botanical nomenclature. When Alexander the Great sent back plant samples from his invasions, Theophrastus documented them and planted cuttings, thereby introducing species such as rice and cotton to Greece.

to know about the world. Aristotle clearly regarded physics—and other science subjects, such as zoology—just as highly as he did philosophical subjects such as ethics and metaphysics. However, at that time, there was still a certain amount of overlap between the subjects, and it was not until the 17th century that philosophy and science were really considered to be distinct.

The father of many sciences

Applying his principle that all findings should be based on observation and reasoning, Aristotle made significant discoveries in a variety of subjects. In astronomy, for example, he disproved the long-held belief that the Earth was flat. Having observed several lunar eclipses, he noted that the Earth’s shadow on the Moon was always curved. This led him to reason, “since it is the interposition of the Earth that makes the eclipse, the form of this line will be caused by the form of the Earth’s surface, which is therefore spherical.”

In the subject of physics, Aristotle laid the groundwork for later advances by Galileo (see pp.54–55) and Newton (see pp.76–81). Although Galileo disproved many of Aristotle’s theories (for example, that the Moon and celestial bodies were perfectly smooth and unchanging), Aristotle’s writings were the starting point for Galileo’s investigations into physics and astronomy. Galileo adapted Aristotle’s principle that force is required



Aristotle’s legacy *endures to this day. His methodical and analytical approach to the pursuit of truth was the starting point for science in its current form, and he believed in learning for its own sake.*